NYU Computer Science Bridge to Tandon Course Winter 2021

Homework 7

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Question 3

Solve the following questions from the Discrete Math zyBook:

a. **Exercise 8.2.2**

Give complete proofs for the growth rates of the polynomials below. You should provide specific values for c and n_0 and prove algebraically that the functions satisfy the definitions for O and Ω.

(b) $f(n) = n^3 + 3n^2 + 4$. Prove that $f = \Theta(n^3)$.

Solution. Let $c = 8$ and $n_0 = 1$, then for any $n \geq n_0$ we have

$$\begin{aligned} n^3 + 3n^2 + 4 &\leq n^3 + 3n^3 + 4n^3 \\ &= 8n^3. \end{aligned}$$

Thus $f(n) = O(n^3)$. Let $c = 1$ and $n_0 = 1$, then for any $n \geq n_0$ we have

$$n^3 + 3n^2 + 4 \geq n^3.$$

Thus $f(n) = \Omega(n^3)$. ■

b. **Exercise 8.3.5**

(a) Describe in English how the sequence of numbers is changed by the algorithm. (Hint: try the algorithm out on a small list of positive and negative numbers with $p = 0$)

Solution. It's a two-pointer algorithm. We have one pointer i starts from the beginning of the sequence while the other pointer j starts from the end of the sequence. They move toward each other until they both meet. The first inner while-loop finds the left-most element greater than or equal to p, and the second inner while-loop finds the right-most element less than p. We then swap these two elements if they exist, and the outer while-loop is executed again. At the end of the algorithm, the sequence would be partitioned into two parts, where the left part are all less than p and the right part are all greater than or equal to p. ■

(b) What is the total number of times that the lines "$i := i + 1$" or "$j := j - 1$" are executed on a sequence of length n? Does your answer depend on the actual values of the numbers in the sequence or just the length of the sequence? If so, describe the inputs that maximize and minimize the number of times the two lines are executed.

Solution. The total number of times that the lines "$i := i + 1$" or "$j := j - 1$" are executed on a sequence of length n is $n - 1$. ■

(c) What is the total number of times that the swap operation is executed? Does your answer depend on the actual values of the numbers in the sequence or just the length of the sequence? If so, describe the inputs that maximize and minimize the number of times the swap is executed.

Solution. The total number of times that the swap operation is executed depends on the actual values of the numbers in the sequence. It is maximized when the numbers in the left half of the sequence are all greater than or equal to p and the numbers in the right half of the sequence are all less than p. In this case, for a sequence of length n, the swap operation is executed $\left\lfloor \frac{n}{2} \right\rfloor$ times. On the other hand, the total number of times that the swap operation is executed is minimized when the numbers in the left part of the sequence are all less than p and the numbers in the right part of the sequence are all greater than or equal to p. Note that the two parts could have different lengths. In this case, the swap operation is executed 0 times. ■

(d) Give an asymptotic lower bound for the time complexity of the algorithm. Is it important to consider the worst-case input in determining an asymptotic lower bound (using Ω) on the time complexity of the algorithm? (Hint: argue that the number of swaps is at most the number of times that i is incremented or j is decremented).

Solution. Given any input of length n, the two inner while-loops will be executed $n - 1$ times in total, and the swap operation is executed at most $\left\lfloor \frac{n}{2} \right\rfloor$ times. Thus the asymptotic lower bound for the time complexity of the algorithm is $\Omega(n)$. Since the algorithm still takes $\Omega(n)$ time even given the best-case input, it is not important to use the worst-case input in determining the time complexity of the algorithm. ■

(e) Give a matching upper bound (using O-notation) for the time complexity of the algorithm.

Solution. By (d), the total number of operations is at most $n - 1 + \left\lfloor \frac{n}{2} \right\rfloor$, which is $O(n)$. ■

Question 4

Solve the following questions from the Discrete Math zyBook:

a) **Exercise 5.1.1**

Consider the following definitions for sets of characters:

- Digits = $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- Letters = $\{\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{e}, \mathrm{f}, \mathrm{g}, \mathrm{h}, \mathrm{i}, \mathrm{j}, \mathrm{k}, \mathrm{l}, \mathrm{m}, \mathrm{n}, \mathrm{o}, \mathrm{p}, \mathrm{q}, \mathrm{r}, \mathrm{s}, \mathrm{t}, \mathrm{u}, \mathrm{v}, \mathrm{w}, \mathrm{x}, \mathrm{y}, \mathrm{z}\}$
- Special characters = $\{*, \&, \$, \#\}$

Compute the number of passwords that satisfy the given constraints.

(b) Strings of length 7, 8, or 9. Characters can be special characters, digits, or letters.

Solution. $40^7 + 40^8 + 40^9$. ■

(c) Strings of length 7, 8, or 9. Characters can be special characters, digits, or letters. The first character cannot be a letter.

Solution. $14 \cdot (40^6 + 40^7 + 40^8)$. ■

b) **Exercise 5.3.2**

(a) How many strings are there over the set $\{\mathrm{a}, \mathrm{b}, \mathrm{c}\}$ that have length 10 in which no two consecutive characters are the same? For example, the string "abcbcbabcb" would count and the strings "abbbcbabcb" and "aacbcbabcb" would not count.

Solution. $3 \cdot 2^9 = 1536$. ■

c) **Exercise 5.3.3**

License plate numbers in a certain state consists of seven characters. The first character is a digit (0 through 9). The next four characters are capital letters (A through Z) and the last two characters are digits. Therefore, a license plate number in this state can be any string of the form:

Digit-Letter-Letter-Letter-Letter-Digit-Digit

(b) How many license plate numbers are possible if no digit appears more than once?

Solution. $10 \cdot 9 \cdot 8 \cdot 26^4$. ■

(c) How many license plate numbers are possible if no digit or letter appears more than once?

Solution. $10 \cdot 9 \cdot 8 \cdot 26 \cdot 25 \cdot 24 \cdot 23$. ■

d) **Exercise 5.2.3**

Let $B = \{0, 1\}$. B^n is the set of binary strings with n bits. Define the set E^n to be the set of binary strings with n bits that have an even number of 1's. Note that zero is an even number, so a string with zero 1's (i.e., a string that is all 0's) has an even number of 1's.

(a) Show a bijection between B^9 and E^{10}. Explain why your function is a bijection.

Solution. Define the function $f : B^9 \to E^{10}$ such that if $x \in B^9$, then $f(x)$ is obtained by first counting the number of 1's in x, and then appending a 1 to x if the number is odd, or appending a 0 otherwise. For example, $f(111110000) = 1111100001$ and $f(111100000) = 1111000000$. f is one-to-one because for x and $y \in B^9$, if $f(x) = f(y)$, then the first nine bits of $f(x)$ and $f(y)$ are clearly the same; that is, $x = y$. f is onto because for any $y \in E^{10}$, the first nine bits $x \in B^9$ and $f(x) = y$. ■

(b) What is $|E^{10}|$?

Solution. Since there is a bijection from B^9 to E^{10}, $|E^{10}| = |B^9| = 2^9$. ■

Question 5

Solve the following questions from the Discrete Math zyBook:

a) **Exercise 5.4.2**

At a certain university in the U.S., all phone numbers are 7-digits long and start with either 824 or 825.

(a) How many different phone numbers are possible?

Solution. $2 \cdot 10^4 = 20000$. ■

(b) How many different phone numbers are there in which the last four digits are all different?

Solution. $2 \cdot 10 \cdot 9 \cdot 8 \cdot 7 = 10080$. ■

b) **Exercise 5.5.3**

How many 10-bit strings are there subject to each of the following restrictions?

(a) No restrictions.

Solution. $2^{10} = 1024$. ■

(b) The string starts with 001.

Solution. $2^7 = 128$. ■

(c) The string starts with 001 or 10.

Solution. $2^7 + 2^8 = 384$. ■

(d) The first two bits are the same as the last two bits.

Solution. $2^8 = 256$. ■

(e) The string has exactly six 0's.

Solution. $\binom{10}{4} = 210$. ■

(f) The string has exactly six 0's and the first bit is 1.

Solution. $\binom{9}{3} = 84$. ■

(g) There is exactly one 1 in the first half and exactly three 1's in the second half.

Solution. $\binom{5}{1}\binom{5}{3} = 50$. ■

c) **Exercise 5.5.5**

(a) There are 30 boys and 35 girls that try out for a chorus. The choir director will select 10 girls and 10 boys from the children trying out. How many ways are there for the choir director to make his selection?

Solution. $\binom{30}{10}\binom{35}{10}$. ■

d) **Exercise 5.5.8**

This question refers to a standard deck of playing cards. If you are unfamiliar with playing cards, there is an explanation in "Probability of an event" section under the heading "Standard playing cards." A five-card hand is just a subset of 5 cards from a deck of 52 cards.

(c) How many five-card hands are made entirely of hearts and diamonds?

Solution. $\binom{26}{5} = 65780$. ■

(d) How many five-card hands have four cards of the same rank?

Solution. $\binom{13}{1} \cdot \binom{48}{1} = 624$. ■

(e) A "full house" is a five-card hand that has two cards of the same rank and three cards of the same rank. For example, queen of hearts, queen of spades, 8 of diamonds, 8 of spades, 8 of clubs. How many five-card hands contain a full house?

Solution. $\binom{13}{1}\binom{4}{3}\binom{12}{1}\binom{4}{2} = 3744$. ■

(f) How many five-card hands do not have any two cards of the same rank?

Solution. $\binom{13}{5} \cdot 4^5$. ■

e) **Exercise 5.6.6**

A country has two political parties, the Demonstrators and the Repudiators. Suppose that the national senate consists of 100 members, 44 of which are Demonstrators and 56 of which are Repudiators.

(a) How many ways are there to select a committee of 10 senate members with the same number of Demonstrators and Repudiators?

Solution. $\binom{44}{5}\binom{56}{5}$. ■

(b) Suppose that each party must select a speaker and a vice speaker. How many ways are there for the two speakers and two vice speakers to be selected?

Solution. $P(44,2) \cdot P(56,2) = 44 \cdot 43 \cdot 56 \cdot 55 = 5827360$. ■

Question 6

Solve the following questions from the Discrete Math zyBook:

a) **Exercise 5.7.2**

A 5-card hand is drawn from a deck of standard playing cards.

(a) How many 5-card hands have at least one club?

Solution. $\binom{52}{5} - \binom{39}{5}$. ■

(b) How many 5-card hands have at least two cards with the same rank?

Solution. $\binom{52}{5} - \binom{13}{5} \cdot 4^5$. ■

b) **Exercise 5.8.4**

20 different comic books will be distributed to five kids.

(a) How many ways are there to distribute the comic books if there are no restrictions on how many go to each kid (other than the fact that all 20 will be given out)?

Solution. 5^{20}. ■

(b) How many ways are there to distribute the comic books if they are divided evenly so that 4 go to each kid?

Solution. $\dfrac{20!}{4!4!4!4!4!}$. ■

Question 7

How many one-to-one functions are there from a set with five elements to sets with the following number of elements?

a) 4

Solution. 0. The is no one-to-one functions since the cardinality of the target is less than that of the domain. ■

b) 5

Solution. $5! = 120$. ■

c) 6

Solution. $P(6, 5) = 720$. ■

d) 7

Solution. $P(7, 5) = 2520$. ■